Adversarial Defense via PGD-Based Adversarial Training

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MSCS2201 Spring 2026

Motivation:
A small perturbation negatively impacts the model accuracy, is there a way to make the model more robust?

Main Idea:
Adversarial training using PGD generated training data.

Result:
PGD Adversarial Training reduces ASR from 100% → 6.69% and achieves 91.72% robust accuracy with <1% clean accuracy loss.

Literature Review

Intriguing Properties of Neural Networks
Szegedy et al. (2014)

First to discover adversarial examples.
Invisible perturbations cause confident misclassification.
Motivated the entire field.

Explaining and Harnessing Adversarial Examples
Goodfellow et al. (2015)

Proposed FGSM — single-step attack.
Vulnerability stems from the linear nature of neural networks.
First basic adversarial training idea.

Towards Deep Learning Models Resistant to Adversarial Attacks
Madry et al. (2018)

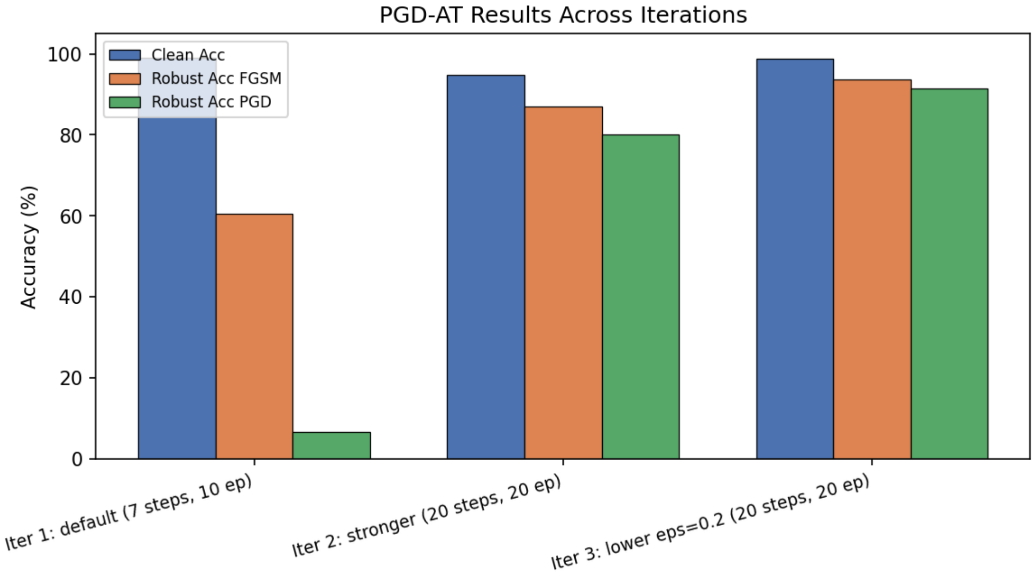
Robustness as min-max optimization.
PGD = strongest first-order attack.
PGD adversarial training is the gold standard defense.

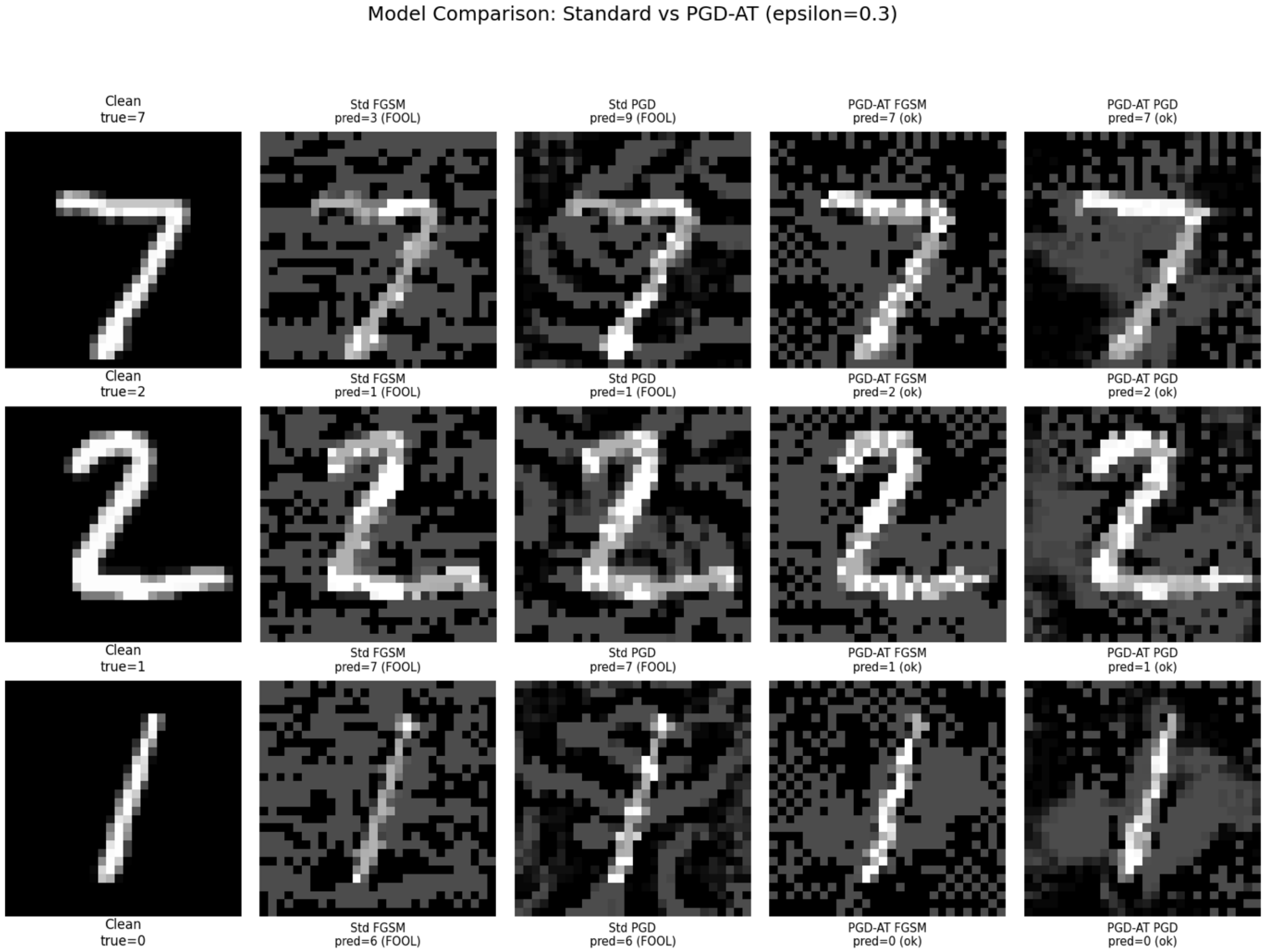
Approach

- Dataset: MNIST
- Model: small CNN
- Eval attacks: FGSM + PGD (40 steps, $\alpha = \varepsilon/10$)
- Optimizer: Adam, lr = 1e-3
- Defense implemented following Madry et al. 2018 min-max PGD adversarial training framework. Attack baselines include FGSM (Goodfellow 2015) and PGD.
- Code: https://github.com/yang330811/ai-mrp-spring-2026-pgd-at

Iteration	ε	PGD train steps	Step size α	Epochs	Notes
iter1	0.3	7	0.01	10	default baseline
iter2	0.3	20	0.03 (ε/10)	20	stronger PGD training
iter3	0.2	20	0.02 (ε/10)	20	smaller perturbation budget

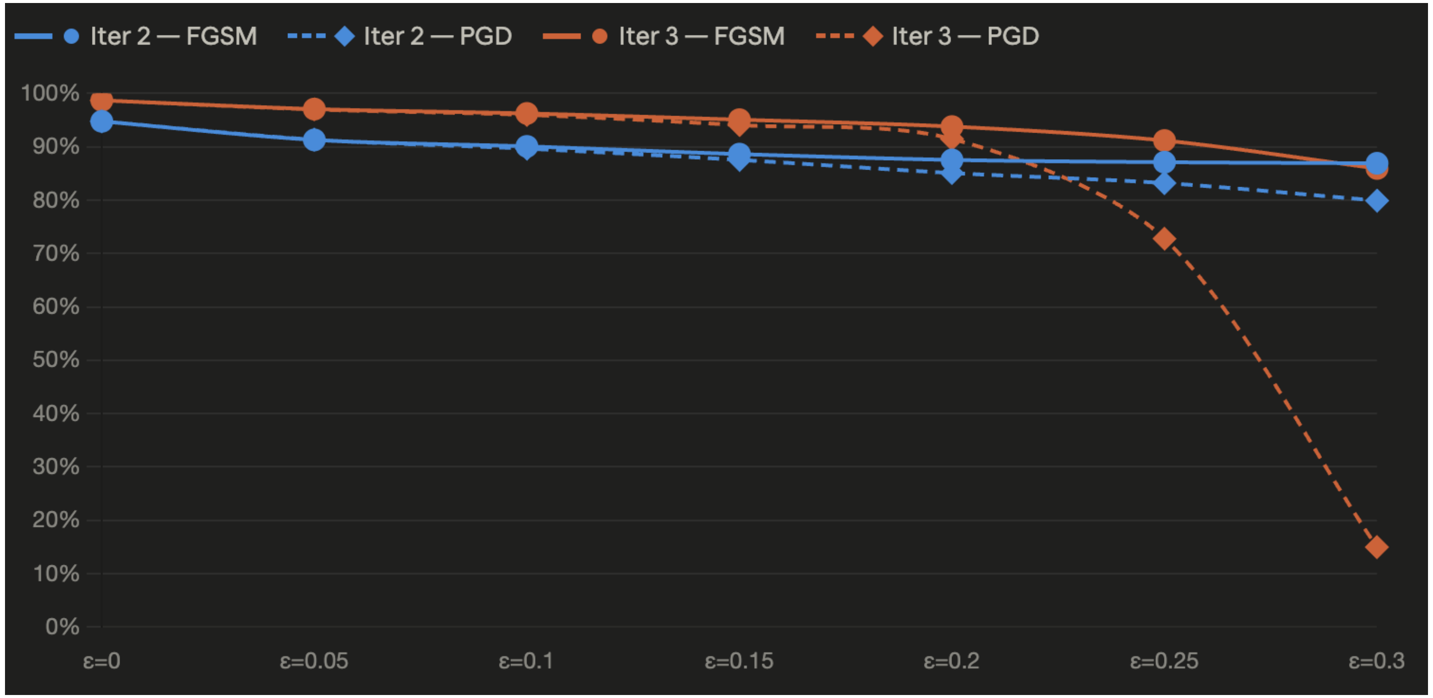
Results





- PGD Adversarial Training reduces Attack Success Rate (ASR) largely from 100% to 6.69%
- More training steps + smaller ε → more robust model
- 91.72% robust accuracy with <1% clean accuracy loss

Results



Iter 2 & 3 : Robustness vs. epsilon comparison

- Models are only robust **within their training ε**
- Iter 3 (ε=0.2) outperforms iter 2 (ε=0.3) within training range, but iter 2 holds up better at high ε
- **Robustness-accuracy trade-off**: stronger training distribution = better defense, but only up to what the model has seen

Takeaway

- PGD-AT is effective but attack-specific — only robust against the attack used in training
- Robustness only holds within training ε — model is vulnerable to stronger out-of-distribution attacks

Demo Video

https://www.youtube.com/watch?v=wVOiLfzmXhE

Reference

- [1] Szegedy, C. et al. (2014). Intriguing properties of neural networks. ICLR 2014. arxiv.org/abs/1312.6199
- [2] Goodfellow, I., Shlens, J., & Szegedy, C. (2015). Explaining and harnessing adversarial examples. ICLR 2015. arxiv.org/abs/1412.6572
- [3] Madry, A. et al. (2018). Towards deep learning models resistant to adversarial attacks. ICLR 2018. arxiv.org/abs/1706.06083